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 **JOHNS HOPKINS**
WHITING SCHOOL
of ENGINEERING

**EN.605.256: Modern
Software Concepts in
Python (Summer 2026)**

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MODULE 3 SQL + FLASK

Grad Cafe Data Analysis

Update Analysis

Pull Data is complete. Click Update Analysis to refresh the query results below.

Database Refresh

Pull Data

Pull Data starts the Module 3 Grad Cafe scraper pipeline, cleans any newly collected records, attempts to refresh LLM fields, and reloads the PostgreSQL applicants table. Update Analysis refreshes this page with the current query results when no data pull is running.

- **Pull Data finished and reloaded 50,019 records into PostgreSQL.**
Downloaded fields and comments were refreshed. Optional local LLM packages are not installed, so deterministic fallback LLM fields were used.
Next step: click Update Analysis to refresh the displayed results.

1

How many entries applied for Fall 2026?**31449**

▼ SQL and explanation

Counts every applicant row whose start term is exactly Fall 2026.

```
SELECT COUNT(*) AS answer FROM applicants WHERE term = 'Fall 2026';
```

2

What percentage of entries are international students?**46.57**

▼ SQL and explanation

Divides entries not marked American or Other by the full table count.

```
SELECT ROUND( 100.0 * COUNT(*) FILTER ( WHERE us_or_international NOT IN ('American', 'Other') ) / NULLIF(COUNT(*), 0), 2 ) AS answer FROM applicants;
```

3

What are the average GPA, GRE, GRE V, and GRE AW values?**GPA: 3.75, GRE Q: 263.61, GRE V: 161.24, GRE AW: 4.34**

▼ SQL and explanation

Uses AVG, which ignores NULL values, so only applicants who provided each metric contribute to that metric.

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS avg_gpa, ROUND(AVG(gre)::numeric, 2) AS avg_gre, ROUND(AVG(gre_v)::numeric, 2) AS avg_gre_v, ROUND(AVG(gre_aw)::numeric, 2) AS avg_gre_aw FROM applicants;
```

4

What is the average GPA of American students in Fall 2026?**3.78****▼ SQL and explanation**

Filters to American Fall 2026 entries and averages non-missing GPA values.

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS answer FROM applicants WHERE term = 'Fall 2026' AND us_or_international = 'American';
```

5

What percent of Fall 2026 entries are acceptances?**37.09****▼ SQL and explanation**

Computes the accepted share among Fall 2026 entries.

```
SELECT ROUND( 100.0 * COUNT(*) FILTER (WHERE status = 'Accepted') /  
NULLIF(COUNT(*), 0), 2 ) AS answer FROM applicants WHERE term = 'Fall 2026';
```

6

What is the average GPA of accepted Fall 2026 applicants?**3.76****▼ SQL and explanation**

Filters to accepted Fall 2026 entries and averages available GPAs.

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS answer FROM applicants WHERE term = 'Fall 2026' AND status = 'Accepted';
```

7

How many entries applied to JHU for a master's degree in Computer Science?**18****▼ SQL and explanation**

Uses the downloaded program string to match JHU or Johns Hopkins, Computer Science, and Masters.

```
SELECT COUNT(*) AS answer FROM applicants WHERE degree ILIKE 'Masters' AND program ILIKE '%Computer Science%' AND ( program ILIKE '%Johns Hopkins%' OR program ILIKE '%JHU%' );
```

8

How many 2026 acceptances applied to Georgetown, MIT, Stanford, or Carnegie Mellon for a CS PhD?**28****▼ SQL and explanation**

Uses downloaded fields to count accepted 2026 CS PhD entries at the requested universities.

```
SELECT COUNT(*) AS answer FROM applicants WHERE term ILIKE '%2026%' AND status = 'Accepted' AND degree ILIKE 'PhD' AND program ILIKE '%Computer Science%' AND ( program ILIKE '%Georgetown University%' OR program ILIKE '%MIT%' OR program ILIKE '%Massachusetts Institute of Technology%' OR program ILIKE '%Stanford University%' OR program ILIKE '%Carnegie Mellon University%' );
```

9

Does question 8 change when using LLM generated fields?**Downloaded fields: 28; LLM fields: 28; result did not change.****► SQL and explanation**

10

Original question: What percentage of entries include applicant comments?**47.71****▼ SQL and explanation**

Measures how often self-submitted rows include non-empty comments.

```
SELECT ROUND( 100.0 * COUNT(*) FILTER (WHERE comments IS NOT NULL AND  
LENGTH(TRIM(comments)) > 0) / NULLIF(COUNT(*), 0), 2 ) AS answer FROM applicants;
```

11

Original question: What are the most common applicant statuses?**Rejected: 21640; Accepted: 19497; Waitlisted: 4835; Interview: 4047****▼ SQL and explanation**

Groups entries by decision status and returns the five most frequent statuses.

```
SELECT status, COUNT(*) AS count FROM applicants GROUP BY status ORDER BY count  
DESC, status LIMIT 5;
```